

Geopolitics and Bitcoin

Team members:

HOU CHONGCONG(侯冲聪)

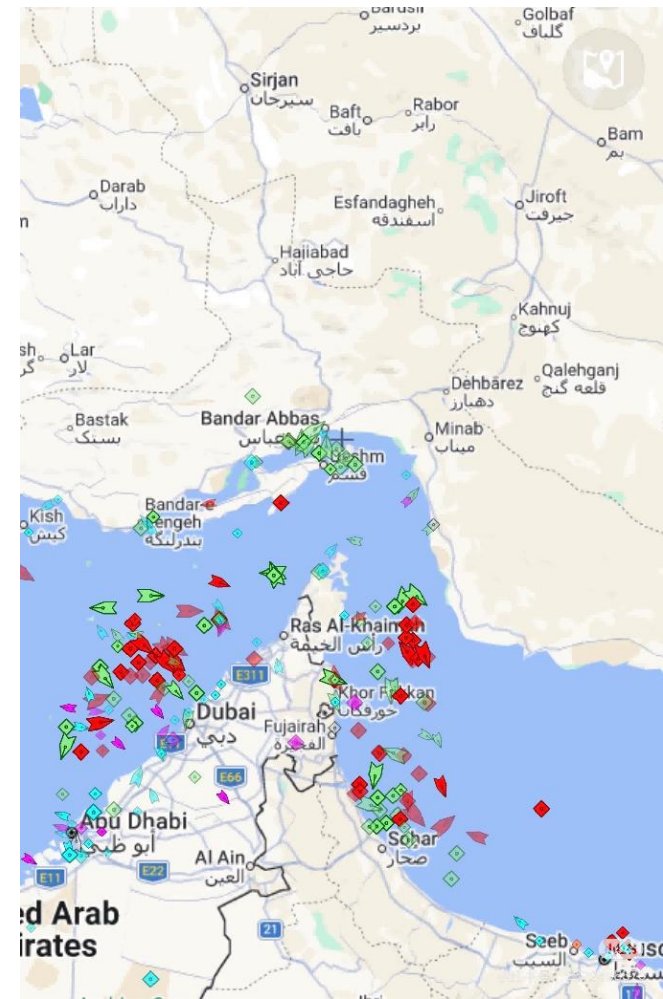
LI XINGLONG(李兴隆)

PENG CONG(彭骢)

LIU YANLING(刘晏伶)

Recent Conflicts

- US attacks Iran
- Strait of Hormuz is closed
- Gasoline prices rises
- Bitcoin prices is influenced



Research Approach

- Correlation Analysis

Method1: LSTM

Method2: Change Point Detection

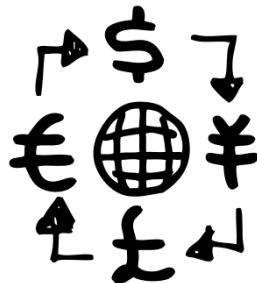
- Mechanism Analysis



Research Approach

Mechanism Analysis

- Potential Path



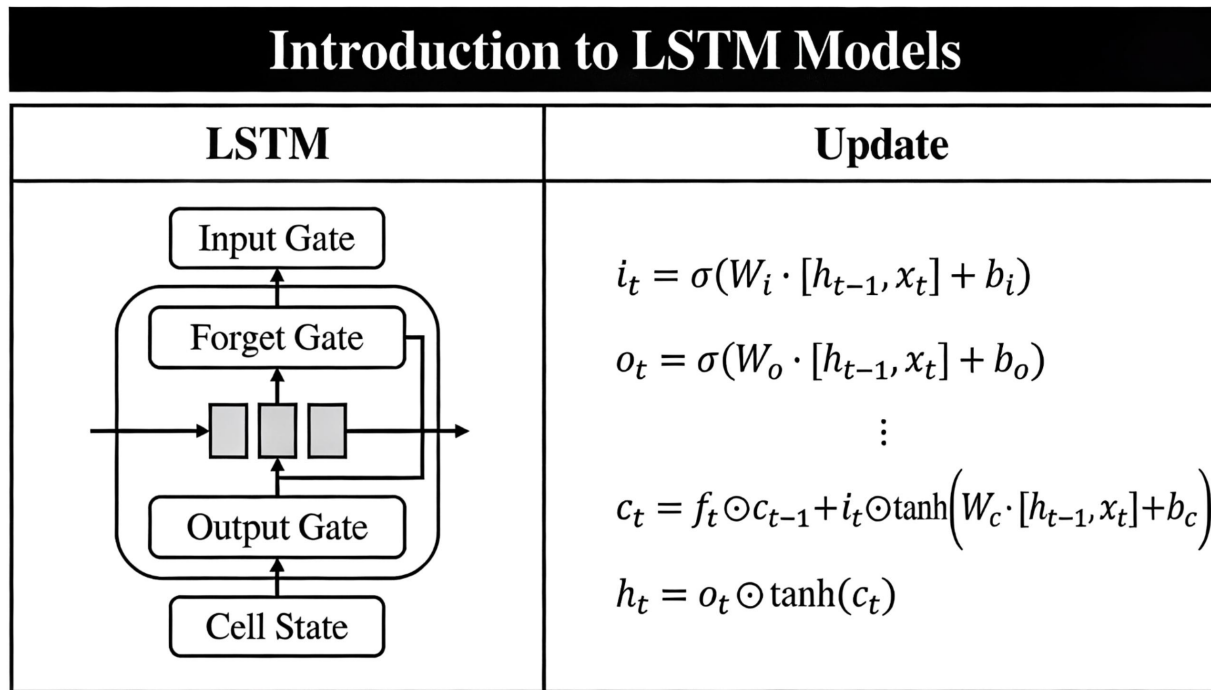
Workflow

- Data cleaning
- Data mining using historical data
- Predicting the future data
- Result analysis and writing



Correlation Analysis

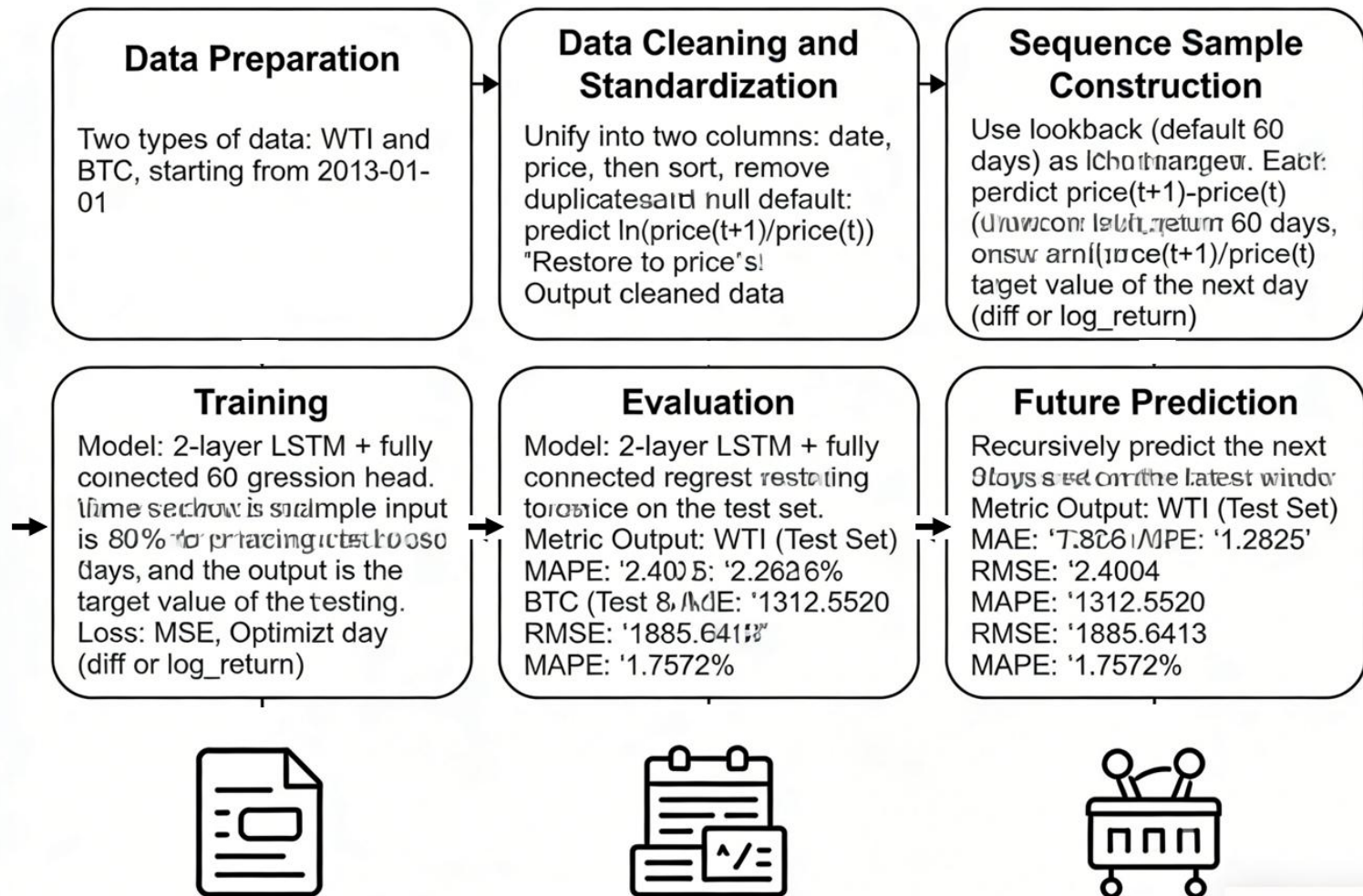
Method1: LSTM



Long Short-Term Memory (LSTM) is a specialized recurrent neural network (RNN) designed to capture long-range dependencies in sequential data. It uses memory cells and three gates—forget, input, and output—to regulate information flow, effectively solving the vanishing gradient problem in traditional RNNs. Widely applied in time series forecasting, natural language processing, speech recognition, and sequential prediction, LSTM excels at retaining critical information over extended sequences.

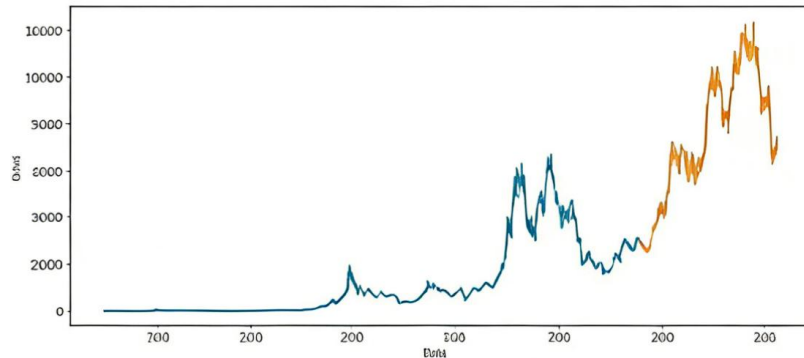
Correlation Analysis 1 --LSTM

Overall Process



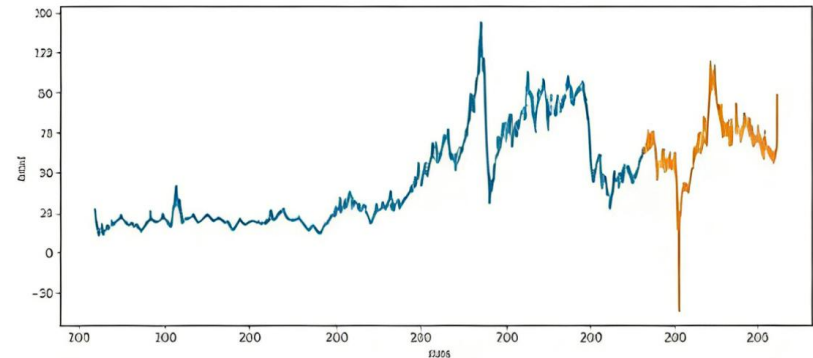
Correlation Analysis 1 --LSTM

LSTM Model Prediction Report: Bitcoin and Crude Oil Price Trends



Bitcoin Price Prediction Analysis (LSTM)

- Historical data range: Starting from 2015 (blue line)
- Test prediction (orange line): 2024-2026, showing significant fluctuations and potential high-level upward trends
- Model evaluation: Demonstrates good trend-capturing ability on the test set.



WTI Crude Oil Price Prediction Analysis (LSTM)

- Historical data range: Starting from 1985 (blue line)
- Test prediction (orange line): 2014-2026, successfully captured the extreme and negative oil prices in 2020.
- Model evaluation: Highly sensitive to fluctuations, predicting price trends.



- Specific points:
 1. The power of the LSTM model has the potential to capture complex patterns in financial and energy asset prices.
- Specific points:
 2. The test prediction results generally match the historical development trend, indicating the model's credibility.
 3. The Bitcoin prediction shows high future volatility and high-level fluctuations, which requires caution. Therefore, emphasizing the importance of risk management is crucial.
 4. Next step: We will focus on demonstrating the ability to capture extreme prices, improving the risk management and assessment of the prediction's uncertainty.
 4. This example has been reserved to train models in the chart, and short-term since the model only shows test predictions, we will continue to improve the model.
 5. Integrate external features: Plan to introduce macroeconomic and policy indicators to improve prediction accuracy.



Correlation Analysis

Method2: Change Point Detection

- **Adaptive algorithm:** a multi-stage iterative algorithm

p--the number of changes in the conditional mean;

q--the number of changes in the conditional variance;

- **References:**

[1] Härdle, W., Werwatz, A., Müller, M., and Stefan, S., 2004, Nonparametric and Semiparametric Models, Springer, Berlin, Heidelberg.

[2] Härdle, 1994, Applied Nonparametric Regression, Cambridge University Press, Berlin, Heidelberg.

[3] Hou, C., C., Wavelet Detection for Change Points in Nonparametric Regression, 2021, Master's thesis, Zhejiang University, China.

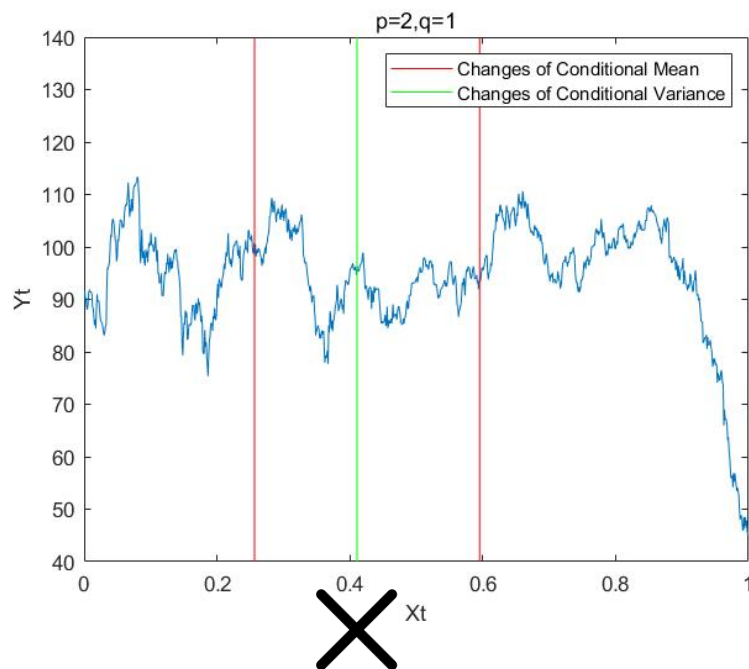


Yeah! This guy is me, and I happen to be the leader of this blockchain project. I used to be a member of Zhang's Lab in School of Mathematical Sciences, Zhejiang University, and we are your fans!

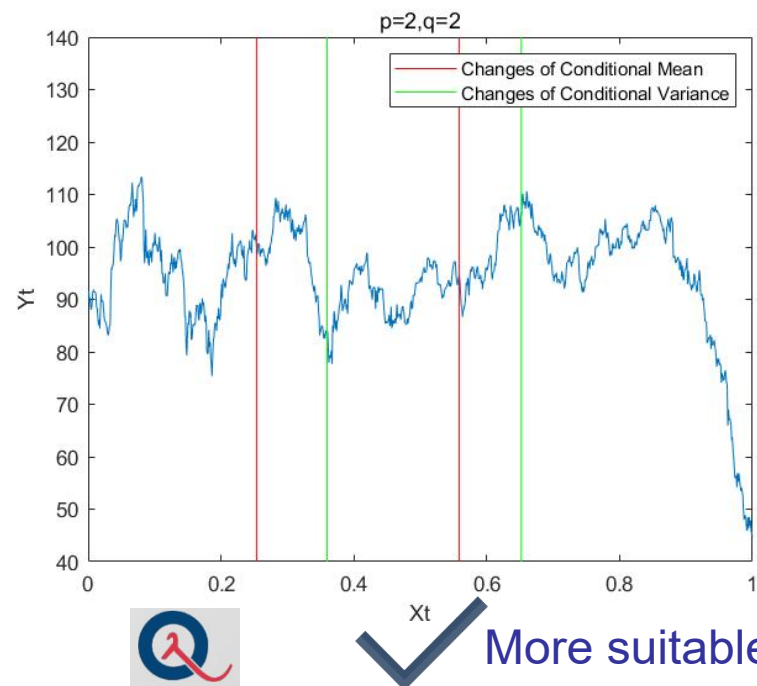
Now I am studying for a PhD in energy economics at XMU. Click my pages! 

Correlation Analysis 2 --Change Point Detection

Crude Oil Price -- daily WTI Index (2013/1/2-2026/3/16)



VS



More suitable !

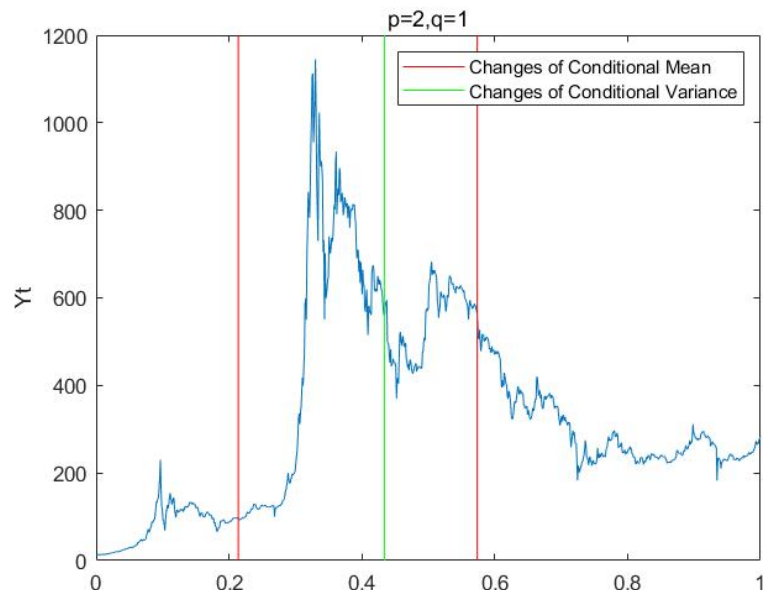
Data Source: U.S. Energy Information Administration

Estimation:

- locations where the conditional mean changes (with corresponding amplitudes):
 $p_1 = 0.2535$ (62.5067) , $p_2 = 0.5584$ (120.025)
- locations where the conditional variance changes (with corresponding amplitudes):
 $q_1 = 0.3596$ (8.7602) , $q_2 = 0.6519$ (6.0465)

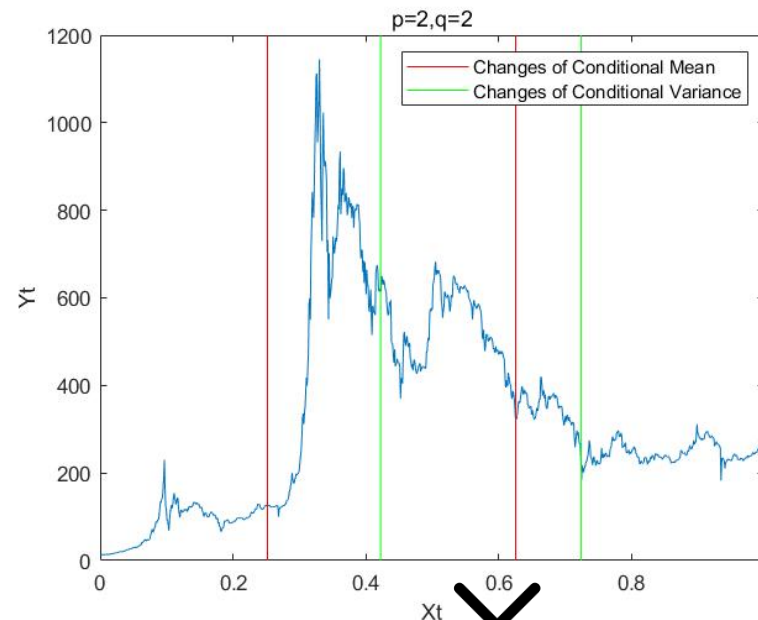
Correlation Analysis 2 --Change Point Detection

Bitcoin Price -- daily closing price (2013/1/1-2026/3/22)



More suitable !

VS



Data Source: Investing.com

Estimation:

- locations where the conditional mean changes (with corresponding amplitudes):
 $p_1 = 0.2140$ (90.7119) , $p_2 = 0.5736$ (90.7119)
- locations where the conditional variance changes (with corresponding amplitudes):
 $q_1 = 0.4342$ (124.7838)

Correlation Analysis 2 --Change Point Detection

- Key Dates of Crude Oil Price

- Abrupt Change in Trend:

- ① 2016/4/29:

Weakening of the US dollar and Influx of speculative funds (2016/4/20-2016/5/10),

Armed Attack in Nigeria (2016/4/20-2016/5/10),

Wildfires in Alberta, Canada(2016/5/1) ⇒ increase in trend

- ② 2020/5/11:

OPEC+largest ever production reduction agreement (2020/5/1)

Passive reduction in shale oil production in the U. S. (2016/5/) ⇒ increase in trend

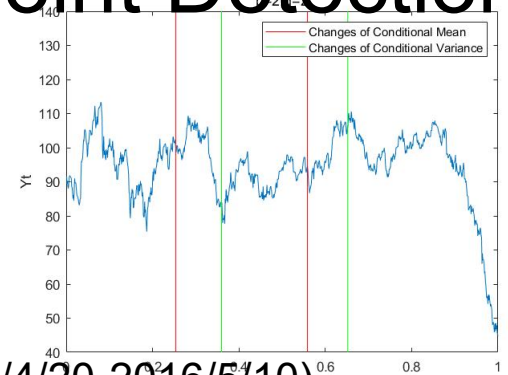
- Abrupt Change in Volatility:

- ① 2017/9/21:

Iraqi Kurdish independence referendum (2017/9/25) ⇒ increase in volatility

- ② 2021/8/3:

Delta pandemic+US dollar strengthens ⇒ increase in volatility



Correlation Analysis 2 --Change Point Detection

- Key Dates of Bitcoin Price

- Abrupt increase in trend:

① 2015/11/2:Regulatory legalization;

Entry of blockchain institutions;

Influx of Chinese funds $\Rightarrow$ increase in trend

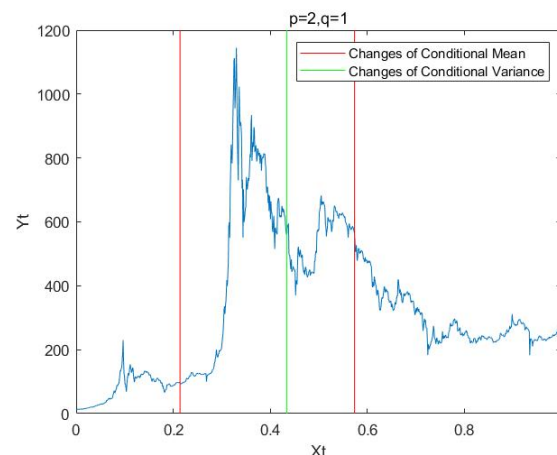
② 2020/8/3:

May be related to abrupt increase in Crude Oil Price (2020/5/11)

- Abrupt increase in volatility:

① 2018/9/30:

Goldman Sachs suspends trading counter plan $\Rightarrow$ increase in volatility



Mechanism Analysis

Test 1: Oil Prices Rise -- → Global Liquidity Tightening

A three-pronged empirical strategy:

Step 1: Mediation Analysis to test the causal chain: Oil Price → Liquidity → Risky Assets

Step2: Structural Break Analysis to examine how the conflict altered the oil-liquidity relationship

Step3: Rolling Window Regressions to capture dynamic coefficient evolution

data

- We use U.S. Dollar Index (DXY) to measure global liquidity.
- Instead of Bitcoin's price, we use Log Returns of Bitcoin to measure devalue.
 - **Statistical Stationarity:** Raw price series are typically non-stationary, exhibiting clear trends and "unit root" characteristics.
 - **Comparability and Scale Independence:** Prices are influenced by an asset's absolute value, while yields reflecting percentage change.
 - Price reflects **cumulative** results, Returns capture **dynamic** transmission.
- Both the DXY and Bitcoin prices is daily data, from 2013-01-01 to 2026-03-28 (without weekends' data since the markets are closed; Inner Merge data processing)

Mechanism Analysis -- Test 1

Step 1: Mediation Analysis

- **Regression Model:** Ordinary Least Squares (OLS)

Regression results:

```
-----
                                OLS Regression Results
=====
Dep. Variable:                liquidity    R-squared:                0.224
Model:                        OLS          Adj. R-squared:           0.224
Method:                        Least Squares    F-statistic:              721.6
Date:                          Sat, 28 Mar 2026    Prob (F-statistic):       7.42e-140
Time:                          23:24:53          Log-Likelihood:           -7266.0
No. Observations:              2500             AIC:                      1.454e+04
Df Residuals:                  2498             BIC:                      1.455e+04
Df Model:                       1
Covariance Type:               nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          89.6664      0.341     263.163      0.000      88.998      90.335
oil_price       0.1375      0.005     26.863      0.000      0.127      0.148
=====
Omnibus:                29.167    Durbin-Watson:           0.014
Prob(Omnibus):           0.000    Jarque-Bera (JB):        18.673
Skew:                   -0.043    Prob(JB):                 8.82e-05
Kurtosis:                2.586    Cond. No.                 256.
=====
```

Results:

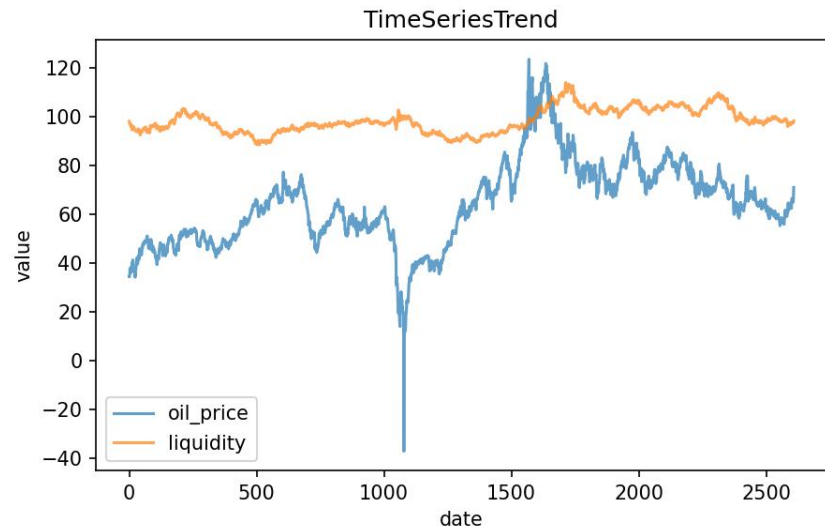
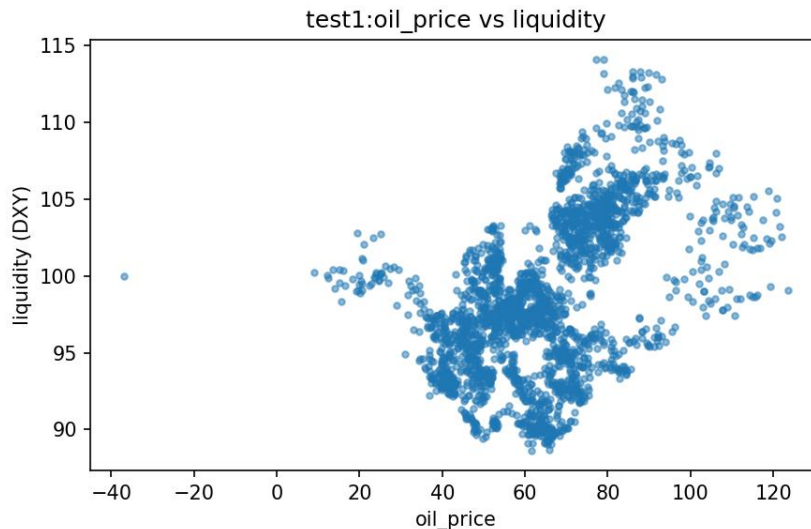
- Oil Price Coefficient=0.137485
- $R^2=0.2241$

Conclusion:

- **✓ Test 1 Passed:** Rising oil prices significantly lead to global liquidity tightening (DXY strengthening).

Mechanism Analysis -- Test 1

Step 1: Mediation Analysis



- **Scatter Plot:** indicating that observations with higher oil prices generally correspond to higher DXY values
- **Time Series:** The two series exhibit periods of synchronized movement, particularly during major geopolitical events (oil supply shocks) and monetary policy shifts



Mechanism Analysis -- Test 1

Step 2: Structural Break Analysis

- To test whether geopolitical conflict amplifies the oil-liquidity relationship,
- **Defined Break Point:** March 2, 2026
- **Period-Wise Regressions:** Estimated oil $\rightarrow$ liquidity coefficients separately for normal and post-conflict periods
- **Interaction Term Model:** $\text{liquidity} = \beta_0 + \beta_1 \times \text{oil} + \beta_2 \times \text{post_iran} + \beta_3 \times (\text{oil} \times \text{post_iran}) + \varepsilon$
- **Chow Test:** Formal test for structural break
- **Rolling Window Regression:** 180-day window to observe coefficient evolution

Mechanism Analysis -- Test 1

Step 2: Structural Break Analysis

Period	Oil Coefficient	Interpretation
Normal Period	0.116054	Baseline impact
Post-Iran Conflict	0.296492	Nearly tripled impact
Additional Effect	0.180438	Conflict-induced amplification

Term	Coefficient	p-value
Oil Price (β_1)	0.116054	0.000
Oil $\times$ Post_Iran (β_3)	0.180438	0.000

- ✓ **Interaction term is significant:** The conflict added an additional 0.1804 to the oil-liquidity coefficient.
- ✓ **Structural break confirmed:** The relationship fundamentally changed after March 2, 2026.

Mechanism Analysis -- Test 1

Step 3: Rolling Window Regression

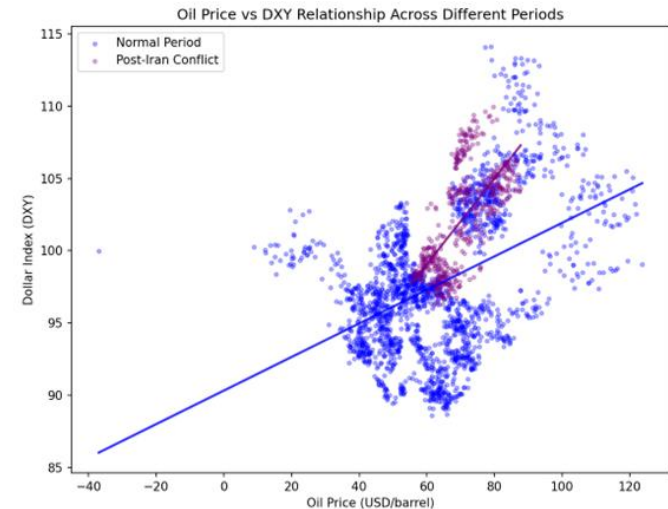
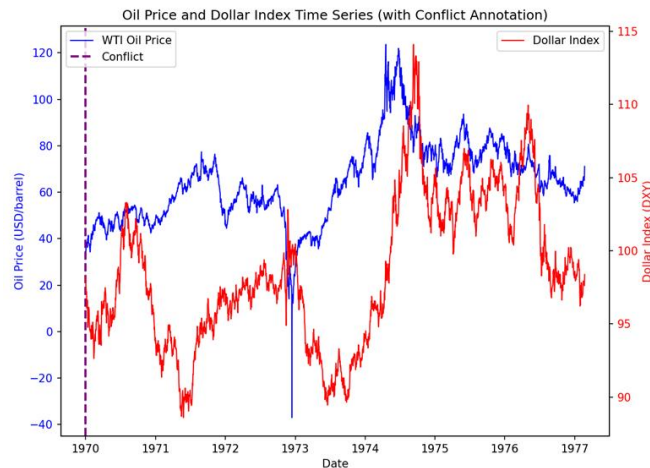
Rolling Window Regression (Window = 180 days)

Pre-conflict average coefficient	0.059808
Post-conflict average coefficient	0.210775
Change magnitude	+252.42%
Coefficient range	-0.4326 ~ 0.6324

- **Interpretation**
- **Normal Period:** Oil price increases have a moderate positive effect on DXY (coefficient = 0.116)
- **Post-Conflict:** Same oil price increase has nearly three times larger effect (coefficient = 0.296)
- **Amplification Mechanism:** Geopolitical risk triggers flight-to-safety, causing markets to react more strongly to oil shocks
- **Dynamic Evidence:** Rolling window shows coefficient rising sharply after the break point, from negative/low values to strongly positive

Mechanism Analysis -- Test 1

Step 3: Rolling Window Regression



Amplified Reactions: Oil price spikes in the post-conflict period are associated with sharper DXY increases

Temporal Alignment: The timing of the structural break aligns with a visible shift in the relationship

Amplification: Post-conflict observations cluster along a steeper line, indicating that markets react more strongly to oil price changes

Robustness: The difference is visually apparent across the entire range of oil prices, not driven by outliers



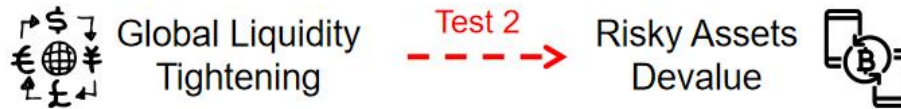
Research Approach

Test 2: Global Liquidity Tightening -- → Risky Assets Devalue

Generally, there is a **negative** correlation between the U.S. Dollar Index (DXY) and Bitcoin (BTC). When the macroeconomic environment shifts toward tightening, the scarcity of the U.S. dollar triggers a flight of capital. As a highly sensitive risk asset lacking intrinsic cash flow support, Bitcoin's "hedging properties" are often overshadowed by intense "liquidity demands" during liquidity crises, leading to significant price declines.

Test strategy		
	Code Tools	Purpose
Step 1: Inflection Point Identification	ruptures (Pelt)	Determine whether the attack constitutes a statistical "mechanism shift."
Step 2: Correlation Analysis	Correlation matrix Heatmap	Find the linear relationship between two factors.
Step 3: LSTM-SHAP	OneHotEncoder & MinMaxScaler TensorFlow/Keras LSTM SHAP	Translate macroeconomic into features recognizable by deep learning; Learn the nonlinear mapping relationship between DXY and BTC. Figure out if the explanatory power of the DXY component for BTC increases or decreases.

Mechanism Analysis -- Test 2



Test Conclusion

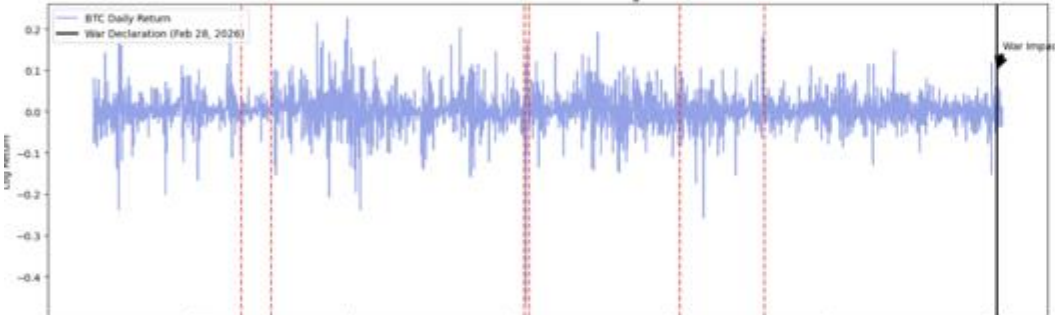
Step 1: Inflection Point Identification

Using the Pelt algorithm to perform a structural analysis of 13 years of historical data

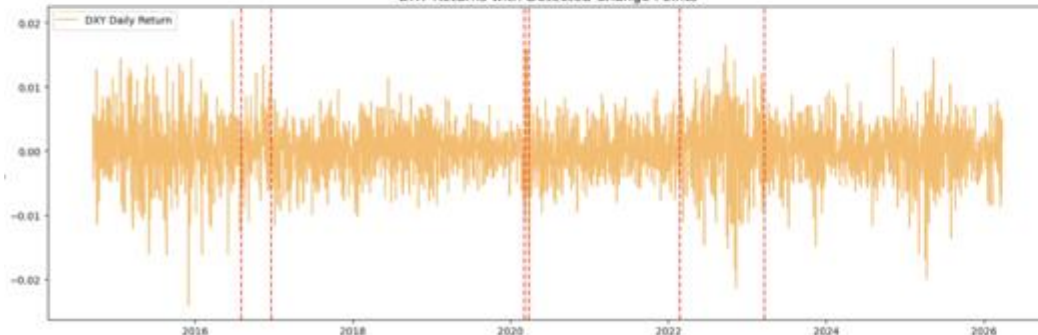
Successfully identified major inflection points such as the 2020 pandemic and the 2022 interest rate hikes, but **failed to detect the point for this event.**

Regarding the impact of the attack, we also failed to detect an inflection point when conducting a localized search starting from January 2025.

BTC Returns with Detected Change Points



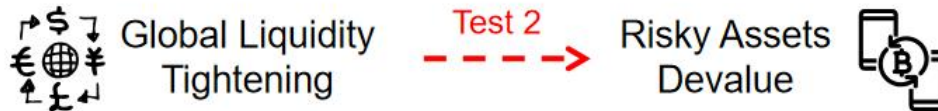
DXY Returns with Detected Change Points



Detected Change Points

- 2016-08-04 Bitcoin Price ↓
Large-scale exchange hacks
- 2016-12-21 ↓
Fed's rate hike cycle and tightening of Chinese regulations
- 2020-03-06 ↓
COVID-19 pandemic - Onset of global stock market panic
- 2020-03-27 ↓
Unlimited quantitative easing policies
- 2022-02-22 ↓
The early stages of the Russia-Ukraine conflict
- 2023-03-21 "safe-haven attributes" has shown
The collapse of Silicon Valley Bank (SVB)

Mechanism Analysis -- Test 2

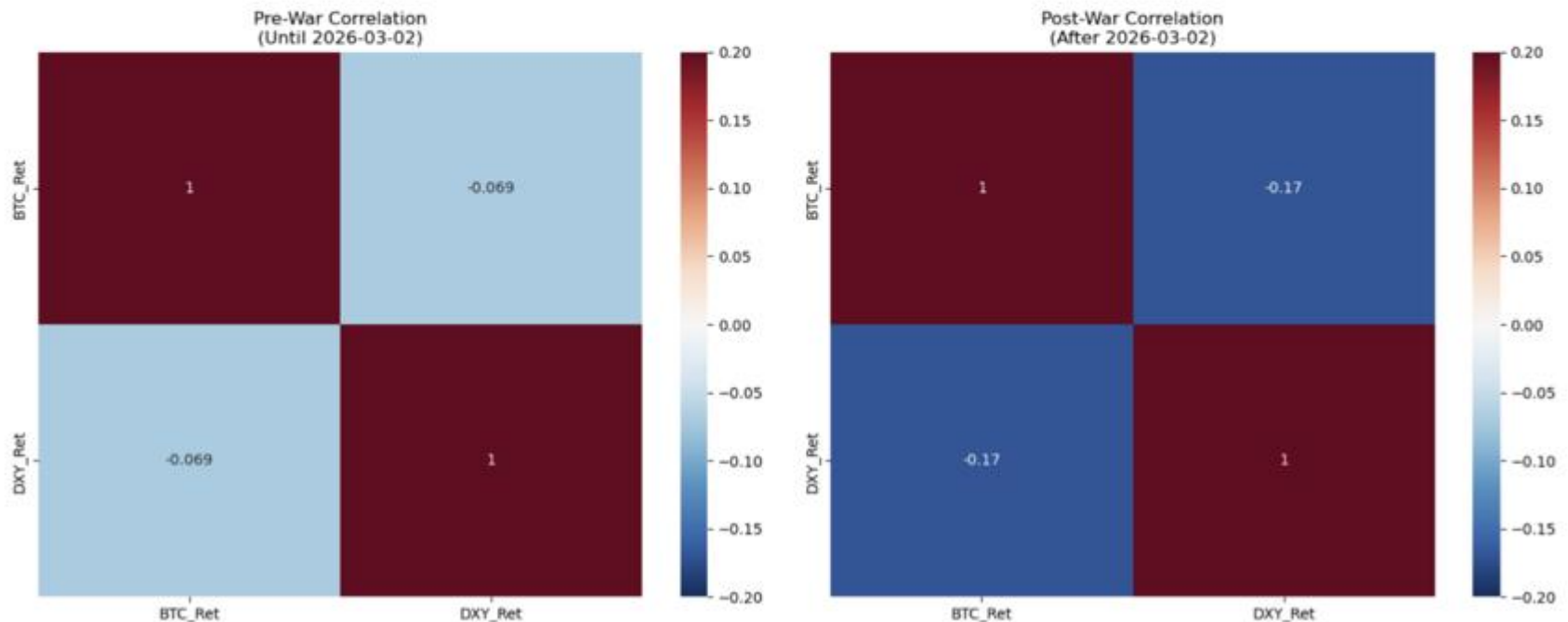


Test Conclusion

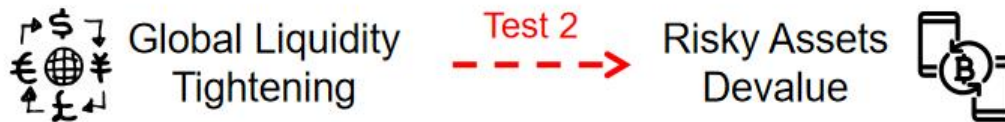
Step 2: Correlation Analysis

Observe Changes in the Absolute Value of Correlation

Following the attack event, the linear correlation between BTC and the DXY strengthened from -0.069 to -0.159, absolute change 0.0998, indicating that the dampening effect of reduced U.S. dollar liquidity on Bitcoin's returns has intensified.



Mechanism Analysis -- Test 2



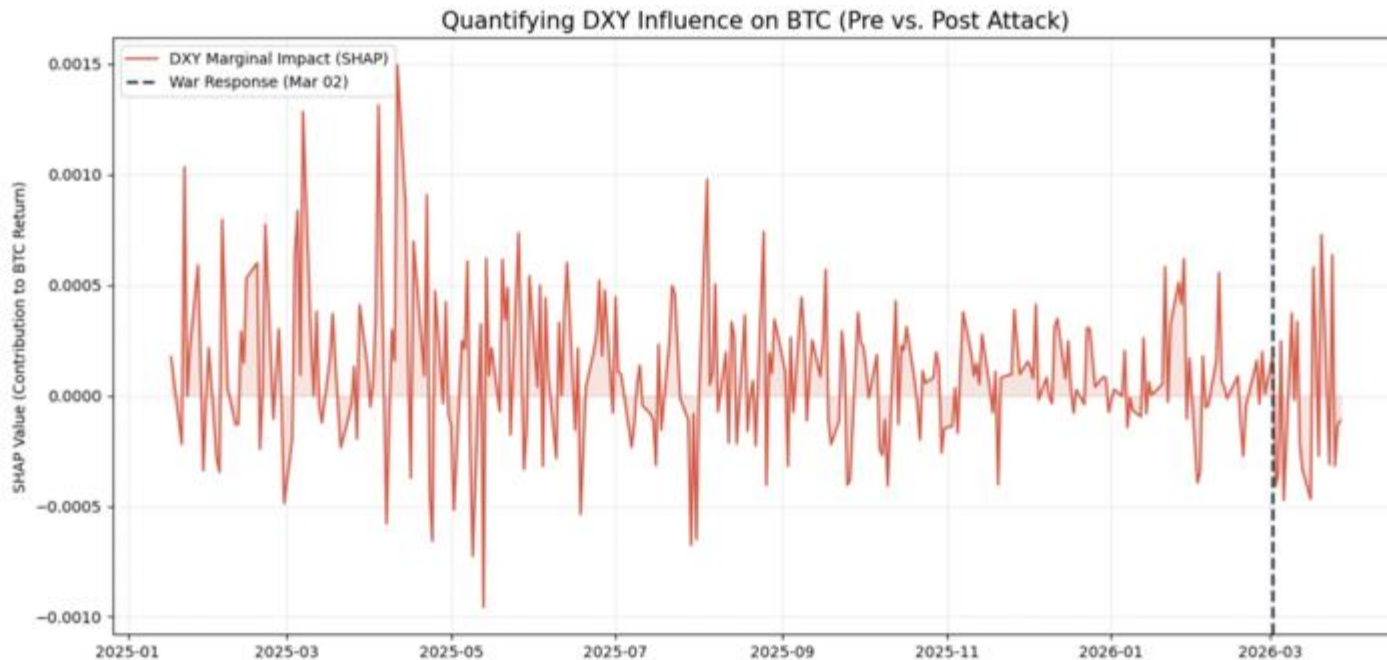
Test Conclusion

Step 3: LSTM-SHAP

Calculate the marginal contribution of the DXY to BTC returns using the GradientExplainer model

SHAP attribution analysis using the LSTM model revealed that after March 2, 2026, the DXY's non-linear contribution actually decreased by **17.60%**.

February 28 is Saturday without DXY data, so set the timepoint at March 2



Mechanism Analysis -- Test 2

Test Conclusion

Linear V.S. Non-linear Relationships

Correlation analysis reveals that a reduction in U.S. dollar liquidity leads to a decline in Bitcoin returns, and this negative correlation intensified following the attack.

However, SHAP attribution analysis using the LSTM model revealed that after March 2, 2026 the nonlinear contribution of the DXY actually decreased by 17.60%

Typically, a strengthening DXY is bearish for Bitcoin. However, under extreme shocks, Bitcoin exhibits clear “desensitization” characteristics. **Market participants may no longer view Bitcoin solely as a “risk asset” but instead begin to activate its “digital gold” safe-haven attributes.**

Consequently, Bitcoin’s pricing logic also shifts from conventional liquidity-driven pricing to pricing driven by abnormal volatility caused by geopolitical risks, thereby weakening the explanatory power of the US Dollar Index .

The study confirms that the attack did not alter the negative correlation between BTC and the DXY, but it significantly weakened the DXY’s predictive efficacy as a leading indicator. This demonstrates that **Bitcoin possesses certain independent pricing attributes in a war-torn environment and no longer relies solely on the US dollar liquidity cycle.**

